

# Student Marketplace App

## Dependency & Setup Report

### 1. Project Overview

The Student Marketplace application is a CRUD-based mobile application developed using React Native with Expo. The app allows authenticated users to create, view, update, and delete marketplace listings stored in Firebase Firestore.

### 2. Development Environment Setup

- 1 Install Node.js and npm
- 2 Create the Expo project using create-expo-app
- 3 Install project dependencies using npm install
- 4 Configure Firebase and Firestore
- 5 Run the development server using Expo

### 3. Main Dependencies

| Dependency                   | Purpose                                                   |
|------------------------------|-----------------------------------------------------------|
| expo                         | Main Expo framework used to build and run the application |
| expo-router                  | File-based navigation and routing                         |
| react-navigation             | Screen and tab navigation                                 |
| react-native-reanimated      | Animations and transitions                                |
| react-native-gesture-handler | Touch gestures and navigation interactions                |
| expo-image                   | Image rendering support                                   |
| expo-haptics                 | Haptic feedback support                                   |
| firebase                     | Authentication and Firestore database services            |

## 4. CRUD Features Implemented

- Create: Users can create new marketplace listings
- Read: Users can browse and view existing listings
- Update: Users can edit their own listings
- Delete: Users can remove their own listings
- Authentication: Firebase Authentication used for login/logout

## 5. Final Notes

The application was developed using React Native, Expo Router, and Firebase Firestore. The project demonstrates full CRUD functionality, navigation, authentication, and cloud database integration.